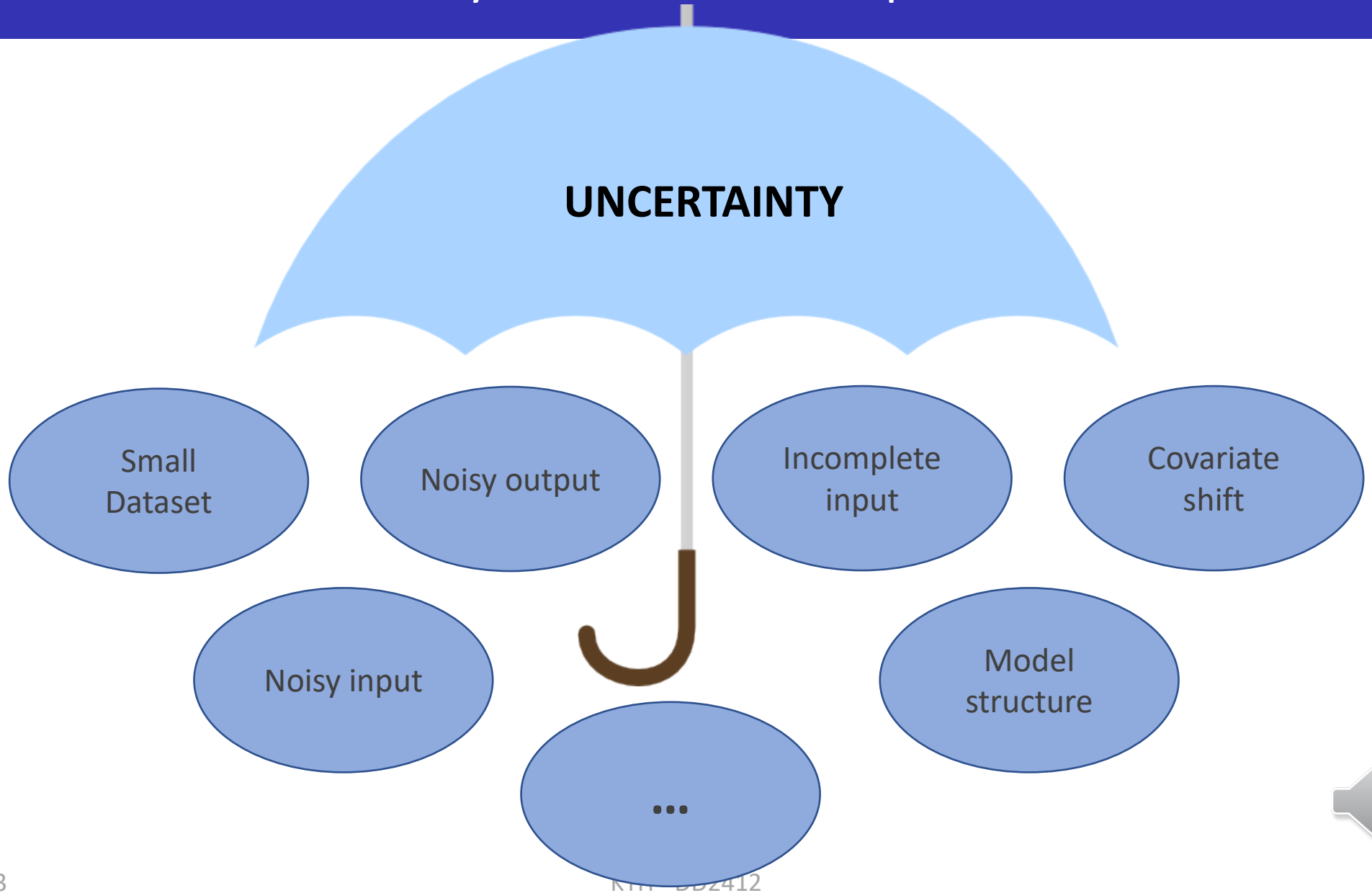
Why do we have predictive uncertainty?



Why do we have predictive uncertainty?



Why do we have predictive uncertainty?

Different types of predictive uncertainty: Epistemic vs. Aleatoric

- Epistemic/Knowledge Uncertainty: lack of model's knowledge
  - Not enough observation
  - Not enough capacity
  - Not a good optimizer
  - Not from the same distribution
  - …
- Aleatoric/Irreducible Uncertainty: unobtainable knowledge! (inherent randomness)
  - Lebel noise
  - Measurement precision
  - Measurement noise
  - Class (definition) overlap
  - …



Reflection/Discussion point

- Can you think of other examples of these two types uncertainty?



3 minutes

- Uncertainty due to hyperparameters, aleatoric or epistemic?

- Epistemic/Model Uncertainty: lack of our (model's) knowledge
  - Not enough observation
  - Not enough capacity
  - …
- Aleatoric/Irreducible Uncertainty: unobtainable knowledge!
  - Lebel noise
    - Measurement precision
    - Class (definition) overlap
  - …

Why do we have predictive uncertainty?

Different types of aleatoric uncertainty

- Homoscedastic Uncertainty
  - if the precision of our ruler is 1mm we always may have a constant uncertainty for each sample

- Heteroscedastic Uncertainty
  - if the precision of our depth-estimation device is dependent on the material we have an input-dependent uncertainty for each sample.



Why do we have predictive uncertainty?

Different types of epistemic uncertainty

- Model Uncertainty
  - epistemic uncertainty due to the *model unable to fit the data.*
    - under-parametrized model e.g., data is more complicated than the model can fit
    - model misalignment e.g., data exhibits forms that the inductive biases of the model prevents it from fitting
    - optimization e.g., non-convergence, bad local minimum, etc.
    - …
- Distributional Uncertainty
  - epistemic uncertainty due to the *lack of observed training data*
    - too sparsely-observed training data
    - distributional shift from the training data to evaluation data
    - …

